Questions by topic: Noise pollution - Set 1 (60 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows typical noise levels of some sounds.

Table 1

Sound	Noise level / dB
Threshold of human hearing (at 1000 Hz)	0
Leaves rustling	20
Library	40
Normal conversation	60
Busy road	80
Chain saw	110
Jet engine at 30 m	140

0 1 . 1 Explain why a logarithmic scale is used to measure sound.

[2 marks]

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0 1 . 2 Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

[2 marks]

Answer ..... times

0 1 . 3 A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

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0 1 . 4 Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

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0 1 . 5 State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

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0 2 . 1 Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

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0 2 . 2 Explain how noise can damage non-living objects.

[3 marks]

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0 2 . 3 Explain how noise can reduce the breeding success of birds.

[2 marks]

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0 2 . 4 Explain why military sonar is a threat to cetaceans.

[2 marks]

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0 3 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 3 . 1 Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 3 . 2 Describe **two** other features of aircraft design that reduce noise.

[2 marks]

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0 3 . 3 Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle .....

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Night flight restrictions .....

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Noise-based landing charges / quota count .....

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Multiple runways .....

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0 3 . 4 Explain why aircraft are noisier on take-off than on landing.

[1 mark]

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0 4 . 1 Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

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0 4 . 2 Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

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0 4 . 3 Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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0 4 . 4 Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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0 5 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 5 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

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0 5 . 2 Explain why readings should be taken with the same wind direction.

[1 mark]

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0 5 . 3 Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

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0 5 . 4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

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0 6 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

Extra space

0 6 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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END OF QUESTIONS

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